

CURRICULUM VITAE

Carlos Pérez-Arancibia

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RESEARCH INTERESTS

Scientific computing; high-order PDE solvers; fast algorithms; numerical analysis; boundary and volume integral equations; wave phenomena; computational electromagnetics; optical metamaterials.

EMPLOYMENT HISTORY

ASSOCIATE PROFESSOR (UHD2, TENURED) Department of Applied Mathematics, Mathematics of Computational Science Faculty of Electrical Engineering, Mathematics and Computer Science University of Twente, Enschede, The Netherlands	<i>Dec 2024 – Present</i>
ASSISTANT PROFESSOR (UD1, TENURED) Department of Applied Mathematics, Mathematics of Computational Science Faculty of Electrical Engineering, Mathematics and Computer Science University of Twente, Enschede, The Netherlands	<i>Sep 2021 – Nov 2024</i>
ASSISTANT PROFESSOR Institute for Mathematical and Computational Engineering Pontificia Universidad Católica de Chile, Santiago, Chile	<i>Jul 2017 – Aug 2021</i>
INSTRUCTOR IN APPLIED MATHEMATICS Department of Mathematics Massachusetts Institute of Technology, Cambridge, MA, USA	<i>Sep 2016 – Jun 2018</i>

EDUCATION

CALIFORNIA INSTITUTE OF TECHNOLOGY, PASADENA, CA, USA · PhD in Applied & Computational Mathematics · Thesis supervisor: Oscar P. Bruno	<i>Aug 2016</i>
PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE, SANTIAGO, CHILE · Diploma in Engineering, Mathematical Engineering (with highest distinction) · Master in Engineering Sciences (with highest distinction) · Minor in Philosophy · Bachelor in Engineering Sciences	<i>May 2010 May 2010 May 2010 Dec 2008</i>
UNIVERSITY TEACHING QUALIFICATION (UTQ) (BASISKWALIFICATIE ONDERWIJS) University of Twente, The Netherlands · Certified according to the Dutch national standards in university-level teaching	<i>Nov 2024</i>

JOURNAL PAPERS¹

- 31.† C. Pérez-Arancibia and C. Turc, *Maxwell à la Helmholtz: Direct boundary integral equations for 3D scattering by perfect electric conductors via Helmholtz operators*, submitted, arXiv:2605.01670, 2026.
- 30.† Y. Tan, C. Pérez-Arancibia, and T. Yin, *A uniformly high-accuracy PML-BIE method for scattering by periodic arrays of obstacles: The 2D case*, submitted, arXiv:2606.07971, 2026.
- 29.† L. M. Faria, C. Pérez-Arancibia and S. Tlupova, *High-order kernel regularization of singular and hypersingular Helmholtz boundary integral operators*, submitted, arXiv:2604.14797, 2026.

¹Papers marked with the symbol † follow the mathematical tradition of alphabetical authorship ordering, whereas those marked with ‡ denote work with mentored students.

- 28.‡ J. Burbano-Gallegos, C. Turc and C. Pérez-Arancibia, *Maxwell à la Helmholtz: Electromagnetic scattering by 3D perfect electric conductors via Helmholtz integral operators*, *ESAIM: Mathematical Modelling and Numerical Analysis*, 60(1):273–315, 2026.
- 27.‡ V. Hojas, C. Pérez-Arancibia and M. A. Sánchez, *Reflectionless discrete perfectly matched layers for higher-order finite difference schemes*, *SIAM J. Sci. Comput.* 46.5 (2024), pp. A3094-A3123.
- 26.† A.-S. Bonnet-Ben Dhia, L. Faria and C. Pérez-Arancibia, *A complex-scaled boundary integral equation for time-harmonic water waves*, *SIAM J. Appl. Math.*, 84.4 (2024), pp. 1532-1556.
- 25.† T. G. Anderson, M. Bonnet, L. M. Faria and C. Pérez-Arancibia, *Fast, high-order accurate numerical evaluation of volume potentials via polynomial density interpolation*, *J. Comput. Phys.*, 511 (2024), p. 11309.
- 24.† T. G. Anderson, M. Bonnet, L. M. Faria, and C. Pérez-Arancibia, *Construction of polynomial particular solutions of linear constant-coefficient partial differential equations*, *Comput. Math. Appl.*, 162 (2024), pp. 94-103.
- 23.† L. Faria, C. Pérez-Arancibia and C. Turc, *Combined field-only boundary integral equations for PEC electromagnetic scattering problem in spherical geometries*, *SIAM J. Appl. Math.*, 84.1 (2024), pp. 1-38.
- 22.‡ T. Strauszer-Caussade, L. Faria, A. Fernandez-Lado and C. Pérez-Arancibia, *Windowed Green function method for wave scattering by periodic arrays of 2D obstacles*, *Stud. Appl. Math.*, 150.1. (2023), pp. 277-315.
- 21.‡ R. Arrieta and C. Pérez-Arancibia, *Windowed Green function MoM for second-kind surface integral equation formulations of layered media electromagnetic scattering problems*, *IEEE Trans. Antennas Propag.*, 70.12 (2022), pp. 11978-11989.
- 20.‡ V. Gómez and C. Pérez-Arancibia, *On the regularization of Cauchy-type integral operators via the density interpolation method and applications*, *Comput. Math. Appl.*, 87 (2021), pp. 108-119.
19. L. Faria, C. Pérez-Arancibia and M. Bonnet, *General-purpose kernel regularization of boundary integral equations via density interpolation*, *Comput. Methods Appl. Mech. Eng.*, 378 (2021), p. 113703.
18. C. Pérez-Arancibia, C. Turc, L. Faria and C. Sideris, *Planewave density interpolation methods for the EFIE on simple and composite surfaces*, *IEEE Trans. Antennas Propag.*, 69.1 (2021), pp. 317-331.
- 17.† D. Nicholls, C. Pérez-Arancibia, and C. Turc, *Sweeping preconditioners for the iterative solution of quasiperiodic Helmholtz transmission problems in layered media*, *J. Sci. Comput.*, 82.44 (2020), pp. 1-45.
- 16.‡ I. Labarca, L. Faria and C. Pérez-Arancibia, *Convolution quadrature methods for time-domain scattering from unbounded penetrable interfaces*, *Proc. R. Soc. A*, 475.2027 (2019), pp. 1-18.
15. C. Pérez-Arancibia, C. Turc and L. Faria, *Planewave density interpolation methods for 3D Helmholtz boundary integral equations*, *SIAM J. Sci. Comput.*, 41.4. (2019), pp. A2065-A2087.
- 14.† C. Pérez-Arancibia, S. Shipman, C. Turc and S. Venakides, *Domain decomposition for quasi-periodic scattering by layered media via robust boundary-integral equations at all frequencies*, *Commun. Comput. Phys.*, 26 (2019), pp. 265-310.
13. C. Pérez-Arancibia, L. Faria and C. Turc, *Harmonic density interpolation methods for high-order evaluation of Laplace layer potentials in 2D and 3D*, *J. Comput. Phys.*, 376 (2019), pp. 411-434.
12. R. Pestourie, C. Pérez-Arancibia, Z. Lin, W. Shin, F. Capasso and S. G. Johnson, *Inverse design of large-area metasurfaces*, *Opt. Express*, 26.26. (2018), pp. 33732-33747.
11. C. Pérez-Arancibia, R. Pestourie and S. G. Johnson, *Sideways adiabaticity: Beyond ray optics for slowly varying metasurfaces*, *Opt. Express*, 26.23. (2018), pp. 30202-30230.
10. C. Pérez-Arancibia, E. Godoy and M. Durán, *Modeling and simulation of an acoustic well stimulation method*, *Wave Motion*, 77 (2018), pp. 214-228.
9. C. Pérez-Arancibia, *A planewave singularity subtraction technique for the classical Dirichlet and Neumann combined field integral equations*, *Appl. Numer. Math.*, 123 (2018), pp. 221-240.
- 8.† C. Jerez-Hanckes, C. Pérez-Arancibia and C. Turc, *Multitrace/singletrace formulations and domain decomposition methods for the solution of Helmholtz transmission problems for bounded composite scatterers*, *J. Comput. Phys.*, 350 (2017), pp. 343-360.

- 7.† O. P. Bruno, E. Garza-Gonzalez and C. Pérez-Arancibia, *Windowed Green Function method for nonuniform open-waveguide problems*, IEEE Trans. Antennas Propag., 65.9 (2017), pp. 4684-4692.
- 6.† O. P. Bruno and C. Pérez-Arancibia, *Windowed Green Function method for the Helmholtz equation in presence of multiply layered media*, Proc. R. Soc. A, 473.2202 (2017), pp. 1-20.
- 5.† O. P. Bruno, M. Lyon, C. Pérez-Arancibia and C. Turc, *Windowed Green Function method for layered-media scattering*, SIAM J. Appl. Math., 76.5 (2016), pp. 1871-1898.
4. C. Pérez-Arancibia and O. Bruno, *High-order integral equation methods for problems of scattering by bumps and cavities on half-planes*, J. Opt. Soc. Am. A, 31.8 (2014), pp. 1738-1746.
3. C. Pérez-Arancibia, P. Zhang, O. P. Bruno and Y. Y. Lau, *Electromagnetic power absorption due to bumps and trenches on flat surfaces*, J. Appl. Phys., 116.12 (2014), p. 124904.
2. C. Pérez-Arancibia, P. Ramaciotti, R. Hein and M. Durán, *Fast multipole boundary element method for the Laplace equation in a locally perturbed half-plane with a Robin boundary condition*, Comput. Methods Appl. Mech. Engrg., 233.1. (2012), pp. 152-163.
1. C. Pérez-Arancibia and M. Durán, *On the Green's function for the Helmholtz operator in an impedance circular cylindrical waveguide*, J. Comput. Appl. Math., 235.1 (2010), pp. 244-262.

CONFERENCE (PEER-REVIEWED) PAPERS²

- L. M. Faria, C. Pérez-Arancibia, and C. Turc. Combined field-only boundary integral equations for electromagnetic scattering. Accepted to *WAVES 2024: The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, June 30–July 5, 2024, Berlin, Germany.
- T. G. Anderson, L. M. Faria, and C. Pérez-Arancibia. Solving boundary and volume integral equations with `Inti.jl`. Accepted to *WAVES 2024: The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, June 30–July 5, 2024, Berlin, Germany.
- T. G. Anderson, M. Bonnet, L. M. Faria, and C. Pérez-Arancibia. Fast, provably high-order accurate methods for volume integral operators. Accepted to *WAVES 2024: The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, June 30–July 5, 2024, Berlin, Germany.
- T. Strauszer-Caussade, L. M. Faria, and C. Pérez-Arancibia, *Windowed Green function method for wave scattering by periodic arrays of 2D obstacles*. Accepted to *WAVES 2024: The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, June 30–July 5, 2024, Berlin, Germany.
- R. Arrieta, L. Faria, C. Pérez-Arancibia, and C. Turc. A high-order density-interpolation-based Nyström method for three-dimensional electromagnetic boundary integral equations. *WAVES 2022: The 15th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, July 24–29 2022, Palaiseau, France.
- J. Hu, E. Garza, C. Pérez-Arancibia and C. Sideris. High-Order accurate integral equation based mode solver for layered nanophotonic waveguides. *International Microwave Symposium*, June 6–11 2021, Atlanta, GA, USA.
- C. Pérez-Arancibia and O. P. Bruno. A high-order integral equation solver for problems of electromagnetic scattering by three-dimensional open surfaces. *WAVES 2015: The 12th International Conference on Mathematical and Numerical Aspects of Wave Propagation*, July 20–24 2015, Karlsruhe, Germany.

SELECTED TALKS AND PRESENTATIONS

INVITED SEMINAR TALKS

- Seminars in PDE and Applications, Delft University of Technology, September 18, 2025.
- POEMS Seminar, ENSTA Paris, Palaiseau, France, July 17, 2025.
- Institute of Computational Mathematics and Scientific Computing Seminar, Chinese Academy of Sciences, May 7, 2024 (online).
- Seminario del Instituto de Ingeniería Matemática y Computacional, Pontificia Universidad Católica de Chile, Santiago, Chile, April 21, 2025.

²The speaker's name is underlined.

- Seminars in Numerical Analysis, Delft University of Technology, Delft, March 16, 2024.
- IEEE GRSS-APS Joint Student Chapter, University of Southern California, Los Angeles, CA, USA, April 7, 2022 (online).
- Applied Mathematics Colloquium, University of Colorado at Boulder, January 21, 2022 (online).
- POEMS Seminar, ENSTA Paris, Palaiseau, France, April 15, 2021 (online).
- Applied Mathematics Colloquium, New Jersey Institute of Technology, Newark, NJ, USA, January 31, 2020.
- Numerical Methods for Partial Differential Equations Seminar, MIT, Cambridge, MA, USA, January 29, 2020.
- Applied Mathematics and Scientific Computing Seminar, Temple University, Philadelphia, PA, USA, January 27, 2020.
- Coloquio del Departamento de Ingeniería Matemática, Universidad de Concepción, Chile, May 23, 2019.
- Mathematical Sciences Colloquium, University of Massachusetts at Lowell, MA, USA, October 13, 2017.
- Seminario del Instituto de Ingeniería Matemática y Computacional, Pontificia Universidad Católica de Chile, Santiago, Chile, August 24, 2017.
- Numerical Methods for Partial Differential Equations Seminar, MIT, Cambridge, MA, USA, April 19, 2017.
- Applied and Computational Mathematics Seminar, University of California, Irvine, CA, USA, February 22, 2016.
- Applied and Computational Mathematics Seminar, University of California, Merced, CA, USA, February 2, 2016.

INVITED MINISYMPOSIUM/WORKSHOP TALKS

- NDNS+ yearly meeting: Nonlinear Dynamics of Natural Systems Cluster, VU Amsterdam, Amsterdam, the Netherlands, March 13, 2026.
- PoWER2025: Propagation of Waves, European Researchers, TU Wien, Vienna, Austria, July 2–4, 2025.
- Institut Mittag-Leffler, Djursholm, Sweden, October 30, 2025. (Invited participant, research program: *Interfaces and Unfitted Discretization Methods*.)
- International Conference on Applied Mathematics (ICAM 2024), City University of Hong Kong, Hong Kong, May 28 – June 1, 2024.
- The 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Tokyo, Japan, August 20–25, 2023.
- Workshop on Computational Methods for Multiple Scattering. Isaac Newton Institute, Cambridge, UK, April 17–21, 2023. [Link to video](#) .
- Conference on Mathematics of Wave Phenomena, Karlsruhe, Germany, February 14–18, 2022 (online).
- Numerical Analysis of Electromagnetic Problems, Oberwolfach Mathematical Research Institute, Germany, March 23, 2021 (online).
- French Latin-American Conference on New Trends in Applied Mathematics, Center for Mathematical Modeling, Universidad de Chile, Santiago, Chile, November 5–8, 2019.
- PUC-Bath Workshop on PDE's and Applications, Santiago, Chile, September 12, 2019.
- SIAM Conference on Computational Science and Engineering, Spokane, Washington, WA, USA, March 1, 2019.
- The 2nd Chilean Symposium on Boundary Element Methods, Universidad Federico Santa María, Valparaíso, Chile, December 14, 2018.
- Caleta Numérica, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile, August 18, 2017.
- The 10th International Conference on Scientific Computing and Applications, Fields Institute, Toronto, Canada, June 6–10, 2016.
- The 13th Annual Conference on Frontiers in Applied and Computational Mathematics (FACM 2016), Newark, NJ, USA, June 3–4, 2016.
- AMMCS-CAIMS Congress, Waterloo, Ontario, Canada, June 7–12, 2015.
- SIAM Conference on Computational Science and Engineering, Salt Lake City, Utah, USA, March 14–18, 2015.
- Valparaíso's Mathematics and its Applications Days, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile, December 12–14, 2012.

CONTRIBUTED TALKS

- The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2024), Berlin, Germany, June 30–July 24, 2024.
- SIAM Conference on Computational Science and Engineering, Amsterdam, The Netherlands, March 1, 2023.

- The 15th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2022), Palaiseau, France, July 25-29, 2022.
- International Conference on Spectral and High Order Methods (ICOSAHOM 2020+1), Vienna, Austria, July 12-16, 2021.
- The 6th Chilean Workshop on Numerical Analysis of Partial Differential Equations (WONAPDE 2019), Concepción, Chile, January 22, 2019.
- The 9th Meeting on Numerical Analysis of Partial Differential Equations (Santiago Numérico III), Santiago, Chile, June 28-30, 2017.
- The 12th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2015), Karlsruhe, Germany, July 20-24, 2015.
- International Conference on Spectral and High Order Methods (ICOSAHOM 2014), Salt Lake City, UT, USA, June 23-27, 2014.

TEACHING EXPERIENCE

UNIVERSITY OF TWENTE

Sep 2021 – Present

Course Instructor

- Analysis I (202200143), 1st term, 2022 and 2023
- Analysis II (202400660), 2nd term, 2022 and 2023
- Introduction to Partial Differential Equations (202400632), 3rd term, 2024 and 2025
- Finite Element Methods: Theory and Applications (202100097), 3rd term, 2024 and 2025
- Calculus A (202001171), 1st term 2025 (~200 students)

PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

Jun 2018 – Aug 2021

Course Instructor

- Calculus III (MAT1630), 1st semester 2020 (~240 students) and 2021 (~190 students)
- Engineering Applications of PDEs and Functional Analysis (IMT3130/3773), 1st semester 2019, 2020, and 2021
- Scientific Computing II (MAT2615), 2nd semester 2020
- Scientific Computing I (MAT2605), 2nd semester 2019
- Advanced Topics in Numerical Analysis (IMT3810), 2nd semester 2019
- Capstone Course on Mathematical and Computational Engineering (IMT3500), 2nd semester 2018

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Sep 2016 – Jun 2018

Course Instructor

- Fast Methods for Partial Differential and Integral Equations (18.336J/6.335J), Fall 2016 and 2017
- Linear Partial Differential Equations: Analysis and Numerics (18.303), Spring 2018

CALIFORNIA INSTITUTE OF TECHNOLOGY

Sep 2012 – Jun 2016

Teaching Assistant

- Methods of Applied Mathematics A (ACM101A), Fall 2014 and 2015
- Methods of Applied Mathematics B (ACM101B), Winter 2015 and 2016
- Introductory Methods of Applied Mathematics A (ACM100A), Fall 2012, 2013, and 2014
- Introductory Methods of Applied Mathematics B (ACM100B), Winter 2013
- Introductory Methods of Applied Mathematics C (ACM100C), Spring 2013, 2014, and 2015
- Introductory Methods of Computational Mathematics B (ACM106B), Winter 2014

PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

Aug 2010 – Dec 2010

Course Instructor

- Mathematical Methods Applied to Engineering (IMM2650), 2nd semester 2010

PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

Mar 2006 – Dec 2009

Teaching Assistant

- Numerical Analysis of Partial Differential Equations, 2nd semester 2009

- Introduction to Numerical Analysis of Partial Differential Equations, 1st semester 2009
- Calculus II, 2nd semester 2008
- Calculus III, 2nd semester 2008
- Partial Differential Equations, 1st semester 2007 and 2008
- Calculus I (Maple laboratory), 1st semester 2008
- Differential Equations, 1st semester 2006 and 2nd semester 2007
- Linear Algebra, 1st and 2nd semesters 2006

FUNDING

- NWO OPEN COMPETITION DOMAIN SCIENCE (M1) awarded by the Dutch Research Council, the Netherlands
 Title: *Density interpolation methods for the fast and high-order evaluation of volume potentials in complex geometries*
 Role: Principal investigator
 Period: 2024 – 2028 (4 years)
 Budget: €372,565
- STRATEGIC RESEARCH INITIATIVE awarded by the 4TU Applied Mathematics Institute, the Netherlands
 Title: *Advancing mathematical methods for wave phenomena*
 Role: Co-principal investigator
 Period: 2024 – 2026 (2 years)
 Budget: €5,000 extendable to €20,000.
- FONDECYT DE INICIACIÓN EN INVESTIGACIÓN awarded by Agencia Nacional de Investigación y Desarrollo, Chile
 Title: *Fast and efficient method of moments for electromagnetic wave propagation and scattering in the presence of unbounded material interfaces*
 Role: Principal investigator
 Period: 2018 – 2021 (3 years)
 Budget: CLP 61,298,000 (~ €75.832)
- MISTI GLOBAL SEED FUNDS awarded by the Massachusetts Institute Technology and the Pontificia Universidad Católica de Chile
 Title: *High-Contrast challenges in numerical wave scattering*
 Role: Co-principal investigator
 Period: 2016 – 2017 (2 years)

AWARDS

- INVITED RESEARCH VISIT, POEMS Lab, ENSTA Paris, Palaiseau, France, Summer 2025 (one month; funded accommodation and travel).
- TOP CHINA UC SANTANDER FELLOWSHIP, December 2018.
- ICES POSTDOCTORAL FELLOWSHIP, UNIVERSITY OF TEXAS AT AUSTIN, February 2016 (declined).
- IMA POSTDOCTORAL FELLOWSHIP, UNIVERSITY OF MINNESOTA, January 2016 (declined).
- PIMS POSTDOCTORAL FELLOWSHIP (CANADA), December 2015 (declined).
- AMMCS-CAIMS STUDENT TRAVEL AWARD, June 2015.
- SIAM STUDENT TRAVEL AWARD, March 2015.
- STUDENT TRAVEL AWARD, NSF Workshop on the BEM, University of Minnesota, April 2012.
- CALTECH INSTITUTE FELLOWSHIP, September 2011.
- CONICYT SCHOLARSHIP FOR MASTER'S STUDIES IN CHILE, January 2009.
- PADRE ALBERTO HURTADO AWARD, Pontificia Universidad Católica de Chile, March 2003.

SERVICE

EDITORIAL ACTIVITIES

- M. Lucido, K. Kobayashi, A. I. Nosich, C. Pérez-Arancibia, and A. Vukovic. Theme issue “Analytically grounded full-wave methods for advances in computational electromagnetics” in *Philosophical Transactions of the Royal Society A* (2025).
- F. Amlani, N. M. Pahlevan, and C. Pérez-Arancibia. Special issue “Recent advances in Fourier-based computational approach for PDEs” in *Advanced Modeling and Simulation in Engineering Sciences*, Springer (2024).

JOURNAL REVIEW

- | | |
|--|---|
| Journal of Computational Physics | SIAM Journal on Applied Mathematics |
| SIAM Journal on Scientific Computing | SIAM Journal on Numerical Analysis |
| Computers and Mathematics with Applications | Advances in Computational Mathematics |
| IMA Journal of Applied Mathematics | IMA Journal of Numerical Analysis |
| • SN Partial Differential Equations and Applications | IEEE Transactions on Antennas and Propagation |
| Engineering Optimization | Int. Journal for Numerical Methods in Engineering |
| Journal of Algorithms and Optimization | Progress in Electromagnetic Research (PIERS) |
| International Journal on Geomathematics | Journal of Applied Mechanics |
| Communications in Computational Physics | Nature Communications |

ORGANIZATION OF SCIENTIFIC EVENTS

- *Wave-Ned 2026: Conference on Analysis, Modeling and Numerics of Waves*. Member of the Organizing Committee (with V. Dolean, A. Geyer, J. Koellermeier, V. Nikolić, L. Scarabosio, and C. Urzúa-Torres), TU Delft, Delft, The Netherlands, August 26–28, 2026.
- *Summer School on Waves: Modeling, Analysis, and Numerics*. Mini-course: “Linear Waves: From Physics to Numerics.” Lecturer and Co-Organizer (with main organizers Vanja Nikolić and Laura Scarabosio), Radboud University, Nijmegen, The Netherlands, August 25–29, 2025.
- Kick-off Workshop of the 4TU.AMI Strategic Research Initiative: *WAVES NL: Advancing Mathematical Methods for Wave Phenomena*. Organizer, University of Twente, The Netherlands, September 20, 2024.
- *Recent Advances on Integral Equation and Spectral Methods for Inhomogeneous Problems* (with Thomas G. Anderson, Rice University). Minisymposium, SIAM CSE 2023, March 2023.
- *Time-Evolution and Frequency-Domain Methods for Partial Differential Equations* (with David Shirokoff, NJIT). Minisymposium, WONAPDE 2019, January 2019.
- Seminar of the Institute for Mathematical and Computational Engineering. Weekly research seminar for graduate and undergraduate students in applied mathematics. Organizer, Pontificia Universidad Católica de Chile, 2020 academic year.
- Numerical Methods for Partial Differential Equations Seminar. Co-organizer together with Manuel A. Sánchez, Pontificia Universidad Católica de Chile, 2nd Semester 2018.

SESSION CHAIR

- The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2024), Berlin, Germany, June 30– July 4, 2024.
- The 15th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2022), Palaiseau, France, July 25–29, 2022.

DEPARTMENT SERVICE

- Faculty Member, Applied Mathematics Programme Committee, Department of Applied Mathematics, University of Twente (Sep 2023 – Apr 2025).
- Module Coordinator, Department of Applied Mathematics, University of Twente (3rd term, 2025).

PARTICIPATION IN PHD COMMITTEES

- Elena Giammatteo, PhD in Applied Mathematics, University of Twente.
- Erli Wind-Andersen, PhD in Mathematical Sciences, New Jersey Institute of Technology.
- Ruben Ailwyn. PhD in Electrical Engineering, PUC Chile.

PARTICIPATION IN MSc COMMITTEES

- Lavinia Lanting, MSc in Applied Mathematics, University of Twente.
- Bert Oudsten, MSc in Applied Mathematics, University of Twente.
- Reinout Nonhebel, MSc in Applied Physics, University of Twente.
- Genaro Laymuns, MSc in Mathematical and Computational Engineering, PUC.
- Pedro Izquiero, MSc in Mathematical and Computational Engineering, PUC.

RESEARCH SUPERVISION

CURRENT PhD STUDENTS

- Juan Burbano Gallegos: University of Twente (Sep 2024 – Present)

FORMER MSc STUDENTS

- Vicente Hojas: PUC Chile (now PhD student, Applied and Computational Mathematics, Caltech)
- Thomas Strauszer: PUC Chile (now PhD student, Applied Mathematics, University College London)
- Rodrigo Arrieta: PUC Chile (now PhD student, Applied Mathematics, MIT)
- Ignacio Labarca: PUC Chile (now PhD in Applied Mathematics, ETH Zürich)

FORMER UNDERGRADUATE STUDENTS AND INTERNS

- Mücahit Özdilekler: Summer Research Intern, Istanbul Technical University
- Robert-Jan Nijhuis: BSc thesis, University of Twente
- Mayank Thakur: BSc thesis, University of Twente
- Gernt Hanskamp: BSc thesis, University of Twente
- Jelle Boon: BSc thesis, University of Twente
- Guilhem Penet: Summer Research Intern, ENSTA Paris, France
- Vicente Gomez: BSc student, PUC Chile (now PhD in Applied Mathematics, NYU Courant Institute)